Reading Notes:
Did Securitization Lead to Lax Screening? Evidence from Subprime Loans
(Keys, Mukherjee, Seru, and Vig, *QJE* 2010)

Notes

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Contents

1 Executive Summary

- **Question:** Did mortgage securitization weaken lenders' screening incentives?
- **Empirical lever:** Investors used a *discrete rule of thumb* around FICO = 620 in the subprime market.
- **Design:** Regression discontinuity (RD) around 620 for *low-documentation* subprime purchase loans.
- **First-stage proxy:** The number of originated/observed low-doc loans jumps sharply at 620 (Table II, Figure II).
- **Core result:** Delinquency *increases* above 620 in low-doc loans (Table III, Figures VI–VII), despite higher FICO.
- **Why this implies lax screening:** Contract terms and observable borrower characteristics are smooth at 620 (Figures III–V, Appendix I.A–I.B), so the performance jump is hard to reconcile with observable selection.
- **Natural experiment:** Anti-predatory lending laws in GA and NJ reduce secondary-market activity; the 620 volume jump shrinks when laws are active (Table IV).
- **Hard-vs-soft info mechanism:** In *full-documentation* loans (hard info disclosed), there is a large volume jump at FICO=600 but no performance deterioration (Table VI; Figures XI–XIII), consistent with the claim that distortions arise when screening relies on soft info.

2 Economic Setting and Mechanism

2.1 Hard information vs. soft information

- **Hard information:** observable/verifiable to outside investors (FICO score, LTV, interest rate, contract terms).
- **Soft information:** costly to produce and not credibly transferable (e.g., qualitative assessment of borrower).

Why focus on low-documentation loans? Low-documentation loans provide less verifiable information to investors, making lender screening based on soft info more important. If securitization weakens screening incentives, the effect should be most visible in low-doc loans.

2.2 Investor heuristics and a discontinuous "ease of securitization"

In the subprime market, participants used discrete cutoffs (notably FICO=620) to summarize credit risk and to structure/price securitizations. The core empirical premise is:

Just-above-620 loans are "easier" to place with investors than just-below-620 loans.

The paper measures this idea via a discontinuity in *the mass of observed loans* at 620.

3 Data and Variables

3.1 Subprime (LoanPerformance) sample

- Home purchase subprime mortgages, vintages 2001–2006.
- Split into:
  - **Low-documentation** purchase loans.
  - **Full-documentation** purchase loans.
- Monthly performance data allow delinquency and prepayment profiles by loan age.

3.2 Outcomes

- **Loan counts by FICO:** number of loans at each integer credit score.
- **Delinquency:** 60+ days delinquent, 90+ days delinquent, foreclosure.
- **Prepayment:** used to check whether differential exit (prepayment) drives apparent default patterns.

3.3 Controls / covariates used for continuity checks

- Contract terms: interest rate, LTV ratio (Appendix I.A, I.C; Figures III–IV).
- ZIP-code demographics from Census 2000: median income, race composition, education, etc. (Appendix I.B, I.C; Figure V).

4 Empirical Design: Regression Discontinuity around FICO = 620

4.1 RD object

Let the running variable be credit score x (FICO). Threshold $c = 620$.

Define:

$$T(x) \equiv \mathbb{1}\{x \geq c\}.$$

The RD estimand (conceptually) is:

$$\Delta_Y(c) \equiv \lim_{x \downarrow c} \mathbb{E}[Y \mid x] - \lim_{x \uparrow c} \mathbb{E}[Y \mid x].$$

4.2 Polynomial RD specification (Equation (1))

For score-level outcomes Y_i (e.g., the number of loans at score i), the paper estimates:

$$Y_i = \alpha + \beta T_i + \theta f(i - 620) + \delta\, T_i \cdot f(i - 620) + \varepsilon_i, \tag{1}$$

where:

- $T_i = \mathbb{1}\{i \geq 620\}$,
- $f(\cdot)$ is a flexible polynomial (implemented as a seventh-order polynomial),
- β is the discontinuity at 620 because all polynomial terms evaluate to 0 at $i = 620$.

4.3 Why β is the discontinuity (a small derivation)

Proposition 1 (Discontinuity interpretation under the specification)**.** *Under* (1)*, the fitted conditional mean at score i is:*

$$\widehat{\mathbb{E}}[Y \mid i] = \widehat{\alpha} + \widehat{\beta} T_i + \widehat{\theta} f(i - 620) + \widehat{\delta} T_i f(i - 620).$$

If $f(0) = 0$, then the fitted jump at the cutoff equals $\widehat{\beta}$.

Proof. Evaluate the fitted mean from the right and left:

$$\widehat{\mathbb{E}}[Y \mid i = 620^+] = \widehat{\alpha} + \widehat{\beta} \cdot 1 + \widehat{\theta} f(0) + \widehat{\delta} \cdot 1 \cdot f(0),$$

$$\widehat{\mathbb{E}}[Y \mid i = 620^-] = \widehat{\alpha} + \widehat{\beta} \cdot 0 + \widehat{\theta} f(0) + \widehat{\delta} \cdot 0 \cdot f(0).$$

Subtracting yields:

$$\widehat{\mathbb{E}}[Y \mid 620^+] - \widehat{\mathbb{E}}[Y \mid 620^-] = \widehat{\beta},$$

because $f(0) = 0$ implies the polynomial terms cancel at the cutoff. □

4.4 Loan-level performance regression (Equation (2))

For delinquency at the loan-month level:

$$Y_{ikt} = \alpha + \beta T_{it} + \gamma' X_{ikt} + \delta' (T_{it} \times X_{ikt}) + \mu_t + \varepsilon_{ikt}, \tag{2}$$

where:

- Y_{ikt} is a delinquency indicator,
- $T_{it} = \mathbb{1}\{\text{FICO}_{it} \geq 620\}$,
- X_{ikt} are observed characteristics (hard information),

- μ_t are time fixed effects,
- interaction terms allow covariate effects to differ on each side of the cutoff.

Interpretation of β. In the baseline without interactions, β captures the difference in delinquency probability (log-odds in logit) for just-above vs just-below 620 loans, conditional on X. With interactions, β is the difference evaluated at the reference values of X; hence marginal effects reported in the paper are the most interpretable objects.

4.5 Placebo-cutoff (permutation) tests

The paper repeatedly estimates RD jumps at many alternative cutoffs and asks whether the true cutoff (620) is an outlier. This helps rule out spurious significance due to flexible polynomials.

5 Evidence Part I: Low-Documentation Loans

5.1 Table I: Summary statistics (what changes over time)

Table I, Panel A (low documentation). Key numbers to internalize:

- **Market growth:** number of low-doc purchase loans rises from 175,882 (2001) to 1,501,840 (2006).
- **Average FICO:** roughly stable, around 633–637.
- **Interest rates:** increase from 8.11% (2001) to 9.28% (2006).
- **Delinquency:** fraction 60+ delinquent rises from 0.06 (2001) to 0.19 (2006).

Table I, Panel B (full documentation).

- Loan counts also expand (290,003 in 2001 to 1,289,420 in 2006).
- Average FICO is lower than in low-doc (about 620–624).
- Delinquency increases over time but remains lower than low-doc (0.04 in 2001 to 0.12 in 2006).

How I use Table I. Table I sets the macro backdrop: (i) the market booms, (ii) underwriting quality deteriorates in later vintages, and (iii) low-doc loans are riskier and more information-problematic.

5.2 Table II + Figure II: First-stage proxy—a mass point at 620

What is Y_i? Here Y_i is the *number of low-documentation loans* at each integer FICO score i. RD in (1) is applied in a 70-point window (FICO 580–649).

Table II results (numbers I should remember). Estimated discontinuity $\widehat{\beta}$ (jump in loan counts at 620):

- 2001: 36.83 (t=2.10) vs mean 117.
- 2002: 72.08 (t=4.10) vs mean 174.
- 2003: 101.70 (t=3.95) vs mean 464.
- 2004: 107.64 (t=5.84) vs mean 742.
- 2005: 85.30 (t=5.20) vs mean 1111.
- 2006: 67.49 (t=2.11) vs mean 1324.
- **Pooled: 80.62 (t=4.20) vs mean 654.**

Interpretation. The mass of originated/observed low-doc loans increases sharply at exactly 620 in every year. This is consistent with the cutoff being economically meaningful for securitization/placement.

5.3 Balance/continuity: Figures III–V and Appendix I.A–I.B

The core RD logic requires that observable characteristics are smooth at 620.

5.3.1 Figures III–IV: Contract terms around 620

- **Figure III: Interest rates** plotted for 615–625 show no visible jump at 620.
- **Figure IV: LTV ratios** plotted for 615–625 show no visible jump at 620.

5.3.2 Figure V: Local income around 620

- **Figure V: Median income** (ZIP-level) plotted for 615–625 is smooth at 620.

5.3.3 Appendix I.A: Regression-based discontinuity in contract terms

Appendix I.A runs RD regressions using *mean interest rate* and *mean LTV* as outcomes.

Key pooled results.

- Interest-rate discontinuity: -0.025 (t=-0.37) vs mean 8.360.
- LTV discontinuity: 0.458 (t=0.60) vs mean 82.782.

Interpretation. Contract terms are economically and statistically smooth at 620.

5.3.4 Appendix I.B: Borrower demographics

Appendix I.B shows that discontinuities in census demographics at 620 are small and statistically insignificant across many variables.

Interpretation. Borrower composition (at least on observables) does not change discretely at the cutoff.

5.4 Performance: Table III and Figures VI–VIII

5.4.1 Table III, Panel A: Annual RD in delinquency (10–15 months after origination)

Outcome: fraction of loans (dollar-weighted) that are 60+ days delinquent between 10 and 15 months after origination.

Estimated jump at 620:

- 2001: 0.005 (t=2.54) vs mean 0.058.
- 2002: 0.010 (t=2.25) vs mean 0.059.
- 2003: 0.014 (t=2.57) vs mean 0.070.
- 2004: 0.019 (t=3.34) vs mean 0.102.
- 2005: 0.024 (t=4.21) vs mean 0.121.
- 2006: 0.044 (t=3.36) vs mean 0.155.

Interpretation. The discontinuity in delinquency is economically meaningful and grows in later vintages.

5.4.2 Table III, Panel B: Pooled robustness across default definitions

Pooled RD jumps at 620:

- 60+ delinquent (dollar-weighted): 0.019 (t=3.32), mean 0.096.
- 90+ delinquent (dollar-weighted): 0.028 (t=4.67), mean 0.076.
- Foreclosure (dollar-weighted): 0.025 (t=6.25), mean 0.031.
- 60+ delinquent (unweighted): 0.025 (t=5.00), mean 0.087.

Interpretation. The performance deterioration at 620 is robust to alternative measures of default severity and weighting.

5.4.3 Table III, Panel C: Loan-level logit evidence

Panel C estimates (2) in a narrow window around 620.

- Baseline (no interactions, no time FE): coefficient 0.12 (t=3.42), marginal effect ≈ 0.004.
- With interactions and time FE: coefficient 0.48 (t=3.33), marginal effect ≈ 0.011.
- Results are similar when clustering by loan and by vintage.

Interpretation. Even conditioning on observed characteristics, just-above-620 loans are more likely to become delinquent.

5.4.4 Figures VI–VIII: Visual evidence and dynamics

- **Figure VI (annual delinquency):** a visible upward shift in delinquency above 620.
- **Figure VII (delinquency by loan age):** the 620+ delinquency curve crosses above the 620- curve after a few months and remains higher.
- **Figure VIII (prepayments):** prepayment profiles are very similar across the cutoff, reducing concerns that differential exit drives delinquency differences.

5.5 Selection concerns and additional checks

5.5.1 Figure IX: Distributions of contract terms

Figure IX compares kernel densities of interest rates and LTV ratios around 620 (example year shown in the paper) and finds them very similar across the cutoff.

Interpretation. There is no evidence that the contract-term distribution shifts discretely and induces a different borrower pool right above 620.

5.5.2 Other confirmatory tests (qualitative summary)

The paper also discusses:

- restrictions to independent lenders,
- arguments that strategic securitization selection would (if anything) bias against finding lax screening,
- additional placebo-cutoff and manipulation tests.

6 Natural Experiment: Anti-Predatory Lending Laws (Tables IV–V)

6.1 Why this helps identification

If the core mechanism is "ease of securitization ⇒ weaker screening", then a shock that reduces secondary-market activity should reduce the discontinuity in loan volumes and weaken the performance reversal.

6.2 Table IV: Low-documentation loans in Georgia and New Jersey

Table IV uses RD around 620 and a policy dummy:

$$\texttt{NoLaw} = \begin{cases} 1 & \text{law not passed or amended (secondary-market activity less constrained)} \\ 0 & \text{law in force (secondary-market activity constrained).} \end{cases}$$

6.2.1 Panel A (number of loans)

- During-law discontinuity is small (e.g., GA: 10.71 vs mean 16).
- When the law is not active, the interaction term is large (GA: +211.50), implying a much larger jump.

Interpretation. The policy shock shifts the "first-stage" discontinuity in volumes, consistent with a securitization channel.

6.2.2 Panel B (delinquency)

The interaction structure implies that the delinquency difference across 620 depends on whether the law is active. The pattern is consistent with delinquency differences weakening when secondary-market constraints bind.

6.3 Table V: Agency (prime) loans

Table V repeats the logic in the agency (prime) market.

6.3.1 Panel A (number of loans)

Small but significant discontinuities in prime-loan counts at 620 exist (e.g., GA: 4.80 vs mean 190.33), but interaction terms with `NoLaw` are not strong.

6.3.2 Panel B (delinquency)

Delinquency rates in agency loans are extremely low (mean around 0.001), and there is no meaningful evidence of a discontinuity pattern that could mimic the subprime results.

Interpretation. The natural-experiment evidence is more consistent with changes in securitization incentives in subprime lending than with broad borrower sorting.

7 Hard-Information Environment: Full-Documentation Loans (Table VI and Figures X–XIII)

7.1 Figure X: Falsification around 620 in full-documentation loans

Figure X examines delinquency by loan age for full-documentation loans around 620. Unlike low-doc loans, there is no performance deterioration above 620, serving as an important falsification.

7.2 Table VI: Full-documentation loans around a different cutoff (600)

Motivation. In full-documentation loans (hard information), the paper uses FICO=600 as a relevant market cutoff and repeats the RD logic.

7.2.1 Panel A (number of loans)

The volume discontinuity at 600 is large:

- pooled jump: 251.83 (t=6.49) vs mean 1020.

7.2.2 Panels B–D (performance)

- Delinquency discontinuities are small and statistically weak (pooled: 0.005 vs mean 0.036).
- Loan-level logit yields a (slightly) negative coefficient at 600 (about -0.06), consistent with better performance above 600 rather than worse.

Interpretation. Large jumps in volume can occur at credit-score cutoffs without generating lax-screening patterns when hard information is already disclosed.

7.3 Figures XI–XIII: Visual confirmation

- **Figure XI:** volume jump at 600 is visually clear.
- **Figure XII:** annual delinquency profiles show no jump at 600.
- **Figure XIII:** delinquency-by-age profiles show no discontinuity at 600.

8 Overall Interpretation and How I would Explain the Paper

8.1 Causal interpretation (what is and is not identified)

- The paper is best viewed as identifying a **reduced-form effect** of crossing a credit-score cutoff that shifts the ease of securitization.
- The empirical evidence supports the **mechanism** that screening distortions are stronger when lending relies on soft information (low-doc), and weaker when hard information is observable (full-doc).
- The design is local to neighborhoods around the cutoff and speaks to marginal loans near FICO 620 (or 600 in the full-doc test).

8.2 Conclusion

Discrete secondary-market heuristics (like the 620 cutoff) can change lenders' incentives in ways that worsen loan quality, especially in environments where screening relies on soft information.